AMSTT

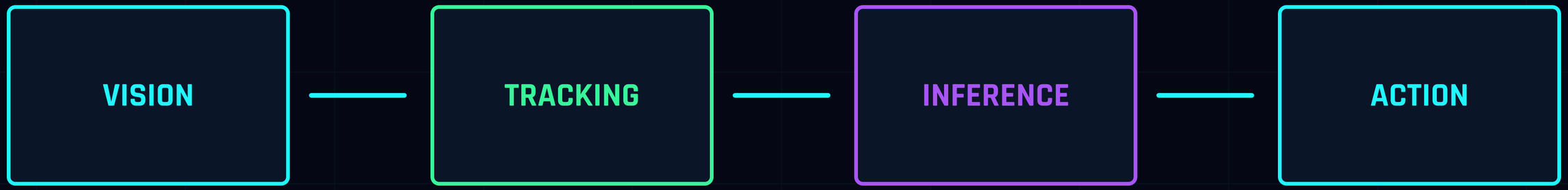
Autonomous Multimodal
Smart Traffic Tracker

Self-driving cars cannot understand human traffic police gestures.

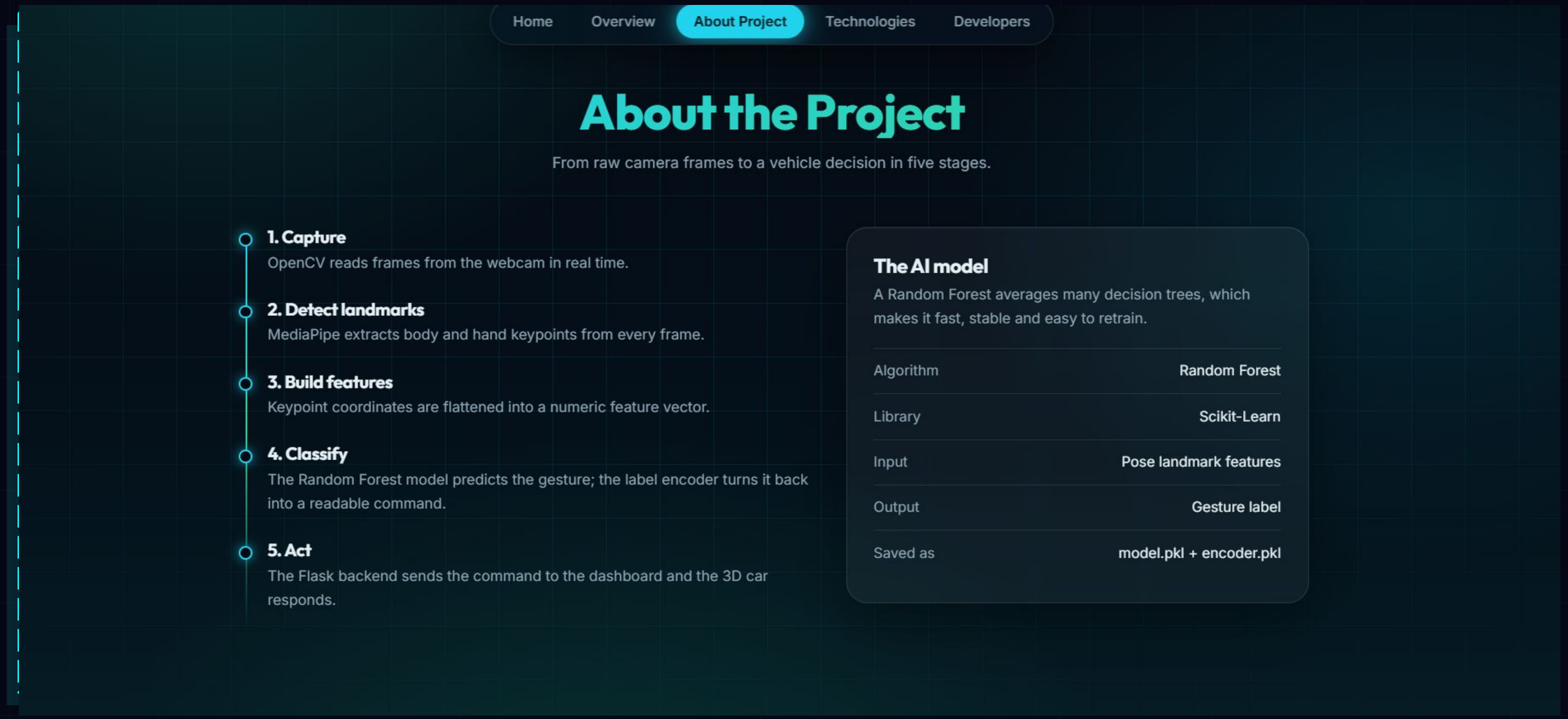
HUMAN SIGNAL ≠ MACHINE ACTION


Bridging human motion
and machine action using AI.

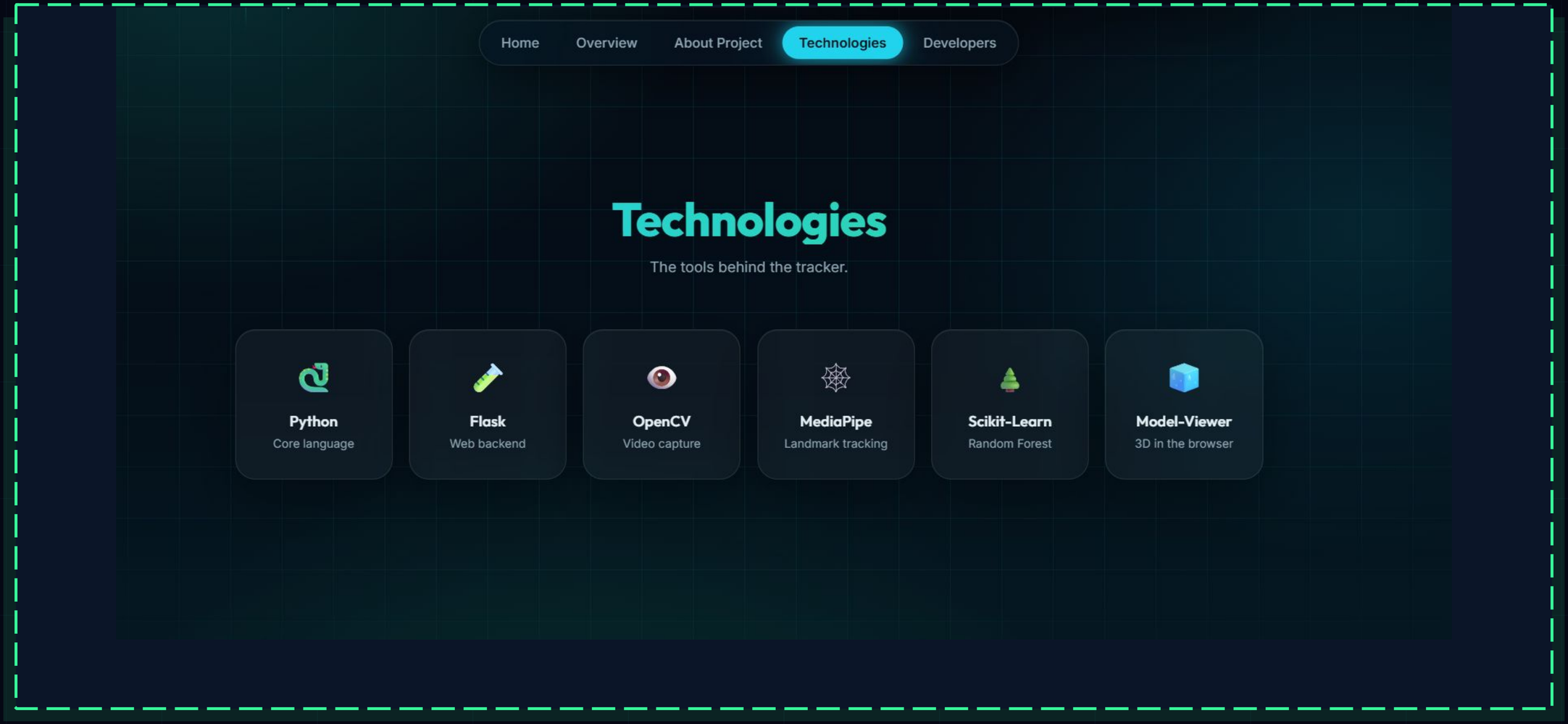
System Architecture



Step 1: Vision Capture



Step 2: Skeletal Tracking



Step 3: AI Inference Engine

AMSTT teaches a simulated vehicle to read a traffic officer's hand signals from a live camera feed and respond without human input.

Gesture Recognition

Hand and body landmarks are tracked frame by frame and classified into signals such as stop, go and turn.

Real-time Processing

A lightweight classifier keeps prediction fast enough to follow a live webcam stream.

Autonomous Vehicles

Each recognised gesture drives the 3D car, so you can see the decision as it happens.

Multimodal Input

Camera video and pose geometry are combined into one feature set for the model.

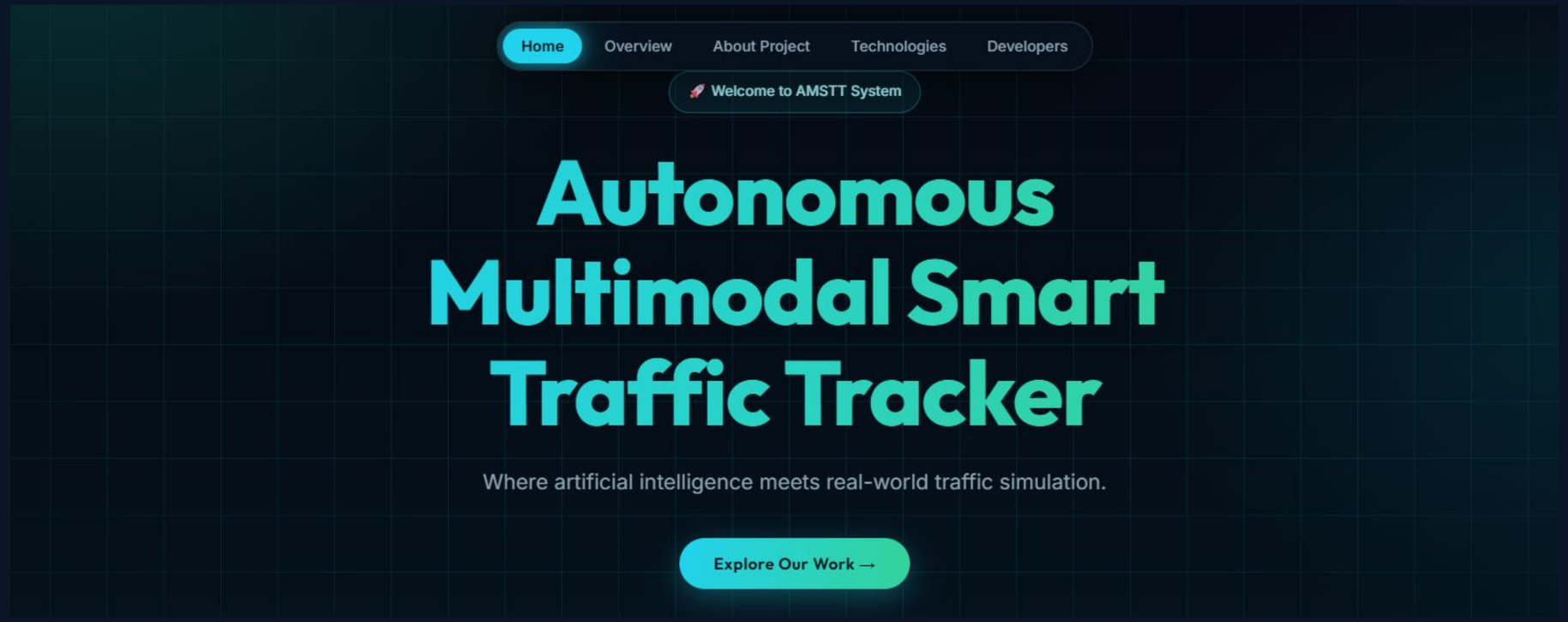
Pluggable Models

Upload any compatible model and encoder file at start-up to swap in a newly trained version.

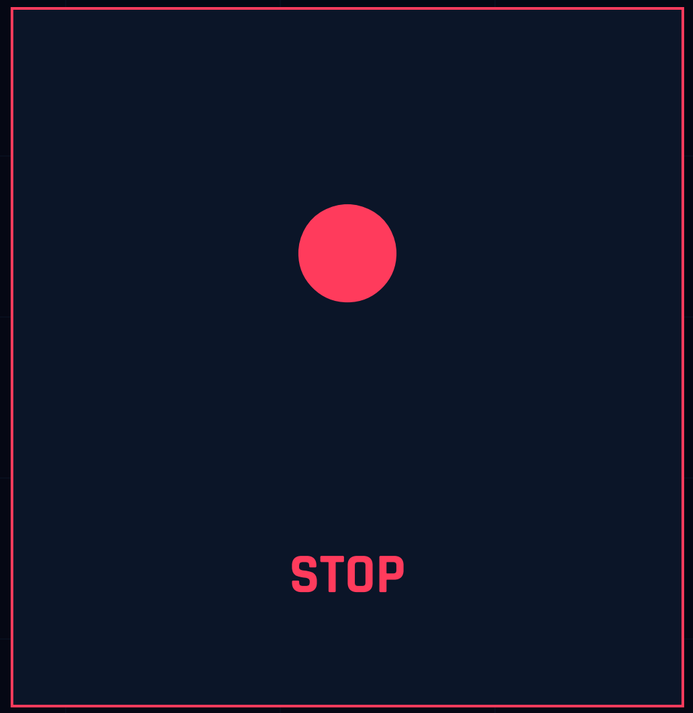
Safer Simulation

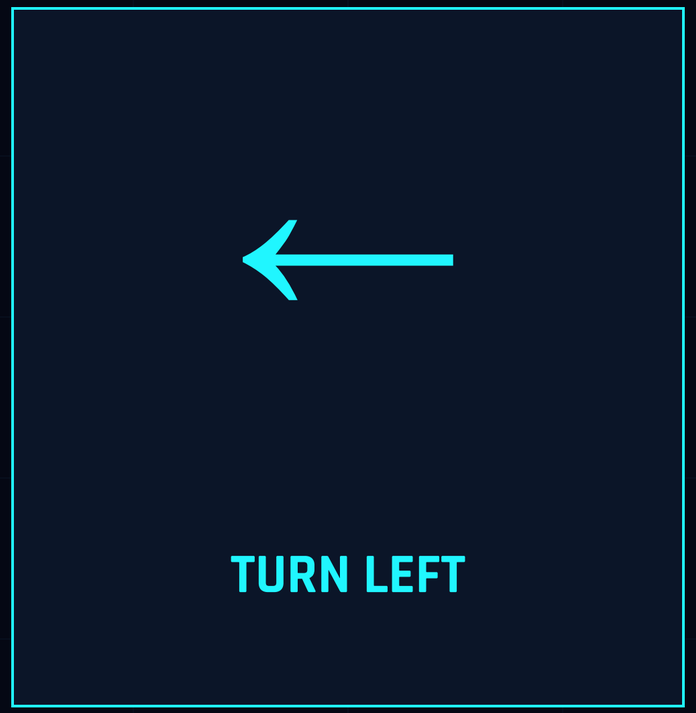
Test traffic-control behaviour on a virtual road before it is ever tried in the real world.

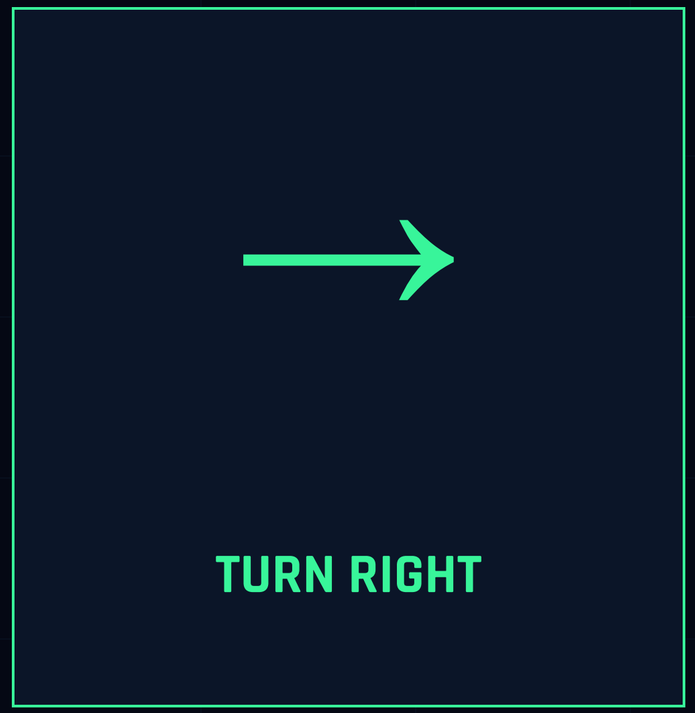
Complete Telemetry Interface



Supported Gestures







Technology Stack

PYTHON

FLASK

OPENCV

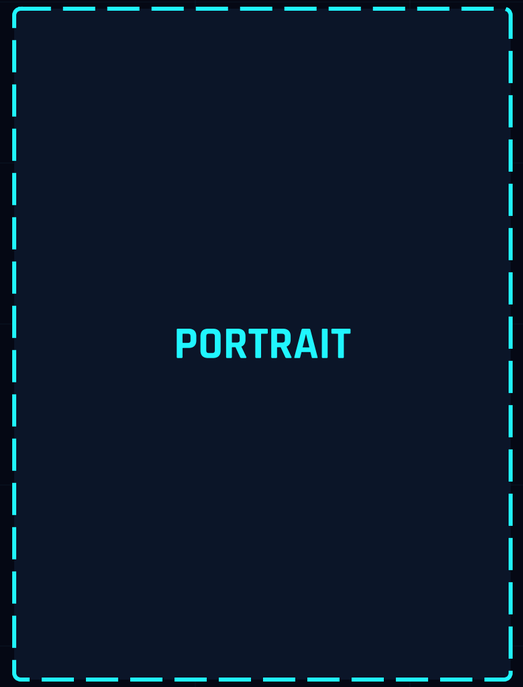
MEDIAPIPE

SCIKIT-LEARN

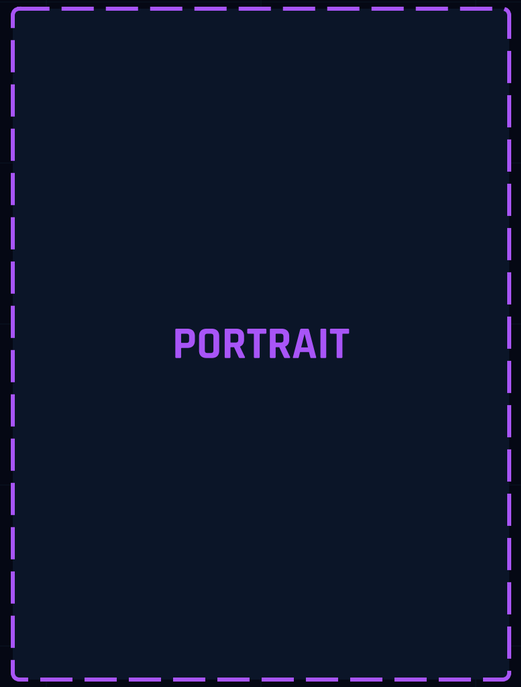
Performance Benchmarks

98%	30	ZERO
ACCURACY	FPS	LAG

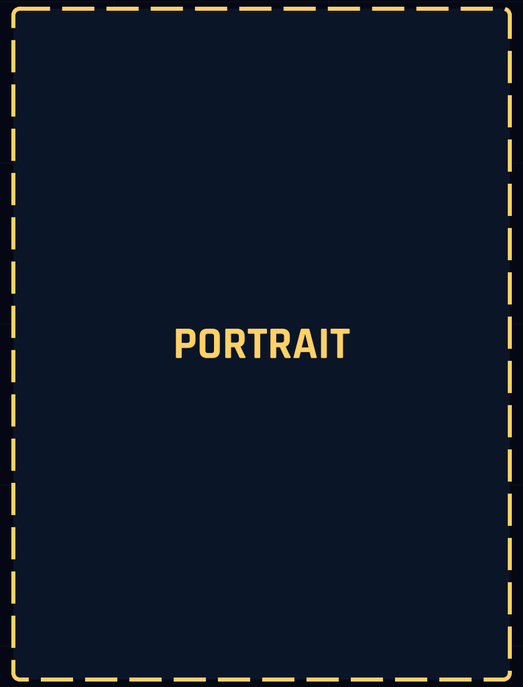
Project Core Team & Mentor



Hariom Patel
Developer



Harsh Mishra
Co-Developer



Dr. Sumit Dhariwal Sir
Project Guide / Mentor

INITIALIZE LIVE DEMONSTRATION

SYSTEM READY